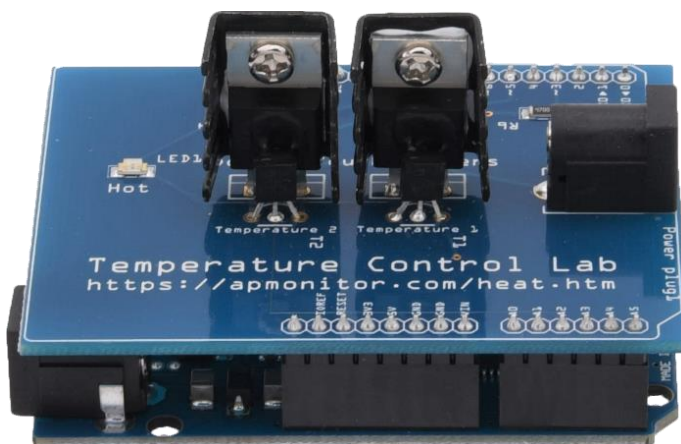


Master in Electrical and Computer Engineering

Master in Aerospace Engineering

Temperature Control Lab

Installation and setup instructions



Prepared by

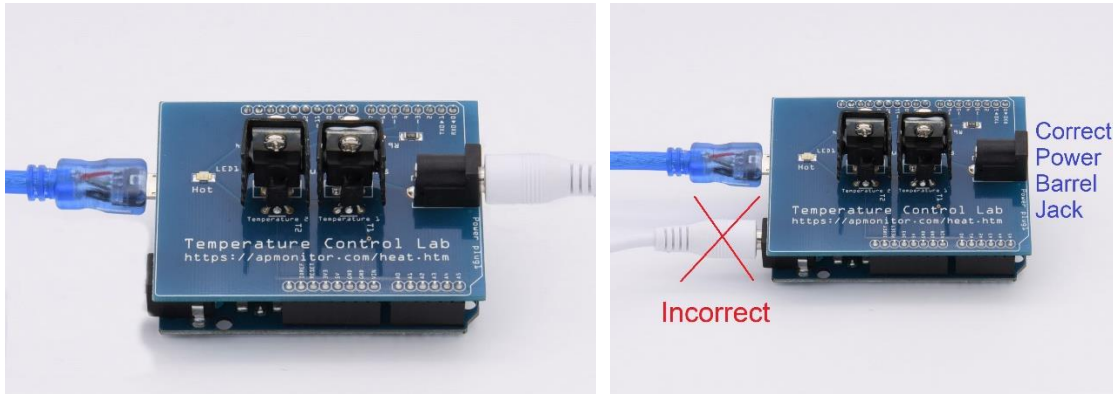
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2022/2023

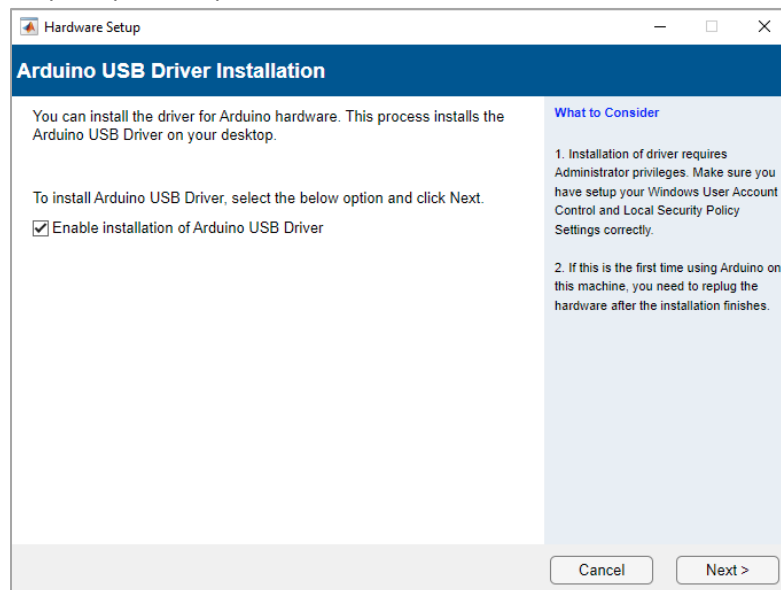
Setup

1. Install MATLAB version R2022b by following the steps in <http://si.tecnico.ulisboa.pt/en/software/matlab/>.
2. Connect the Arduino Leonardo to your PC with the Micro-USB interface.
3. Power the TCLab shield using the power plug of the upper board, not the Arduino board, as pictured below.



4. Install the MATLAB Support Package for Arduino Hardware from <https://www.mathworks.com/hardware-support/arduino-matlab.html>, or from within MATLAB in the Add-On Explorer  in the “Home” tab. After the installation, select “Setup Now”.

- 4.1. In the Hardware Setup, enable the “Enable installation of Arduino USB Driver” box and click “Next”. If prompted for permissions, allow them.



- 4.2. When prompted with “Do you want to set up your Arduino board connection” select “Yes” and click “Next”.

Arduino Hardware Setup

Do you want to set up your Arduino board connection?

☒ Yes

☐ No

4.3. Choose connection type “USB” and click “Next”

Choose Connection Type

Connect Arduino via USB and choose your connection type. Supported types:

☒ USB

☐ Bluetooth®

☐ WiFi

4.4. Choose the “Leonardo” board and the port “COM3” from the drop-down menus. All libraries may be disabled. Click “Program” and ensure the upload is successful. Click “Next”.

Upload Arduino Server

Connect Arduino to host computer via USB

Choose board: Leonardo Choose port: COM3

Select libraries to be included in the server:

☒ Select All

☐ APDS9960

☐ Adafruit/MotorShieldV2

To begin uploading server to the board, click 'Program'.

Program

Success! Click Next to proceed.

4.5. Test the board connection and ensure it is successful. Click “Next” and then “Finish”.

Test Arduino Connection

Current Settings:

Connection Type	USB
Port	COM3
Board	Leonardo
Libraries	

Test connection

☒ Test connection successful. To use your Arduino board with the current settings, use the Arduino Explorer app or type a = arduino. For unofficial (clone) Arduino boards, type a = arduino(port.board).

If necessary, you can return to the setup page later by searching for the add-on in the list of installed add-ons in the Add-On Manager and clicking the  button on the right.

5. Install the Simulink Support Package for Arduino Hardware from <https://www.mathworks.com/hardware-support/arduino-simulink.html> or from within MATLAB with the Add-On Explorer. After the installation, select “Setup Now”.

5.1. In the Hardware Setup, select the Arduino Leonardo and click “Next”.

Select Hardware

Select Hardware from the list:

Arduino Hardware:

☐	Select All
☐	Arduino Uno
☐	Arduino Mega 2560
☐	Arduino Mega ADK
☒	Arduino Leonardo
☐	Arduino Micro
☐	Arduino Robot Motor ...
☐	Arduino Robot Control...
☐	Arduino Due Program...
☐	MKR1000
☐	MKRZero
☐	MKR WiFi 1010

Arduino Compatible Hardware:

☐	Select All
☐	ESP32 Wroom

About Your Selection
Pre-selected Arduino hardware work with the support package. Selecting Arduino Compatible hardware require upgrading the required third-party software. A pre-selected Arduino Compatible hardware indicates the presence of required third-party software for its usage.

What to Consider
The hardware set-up process for Arduino and Arduino compatible hardware consists of setting up the right third-party software for smoother development and deployment of applications

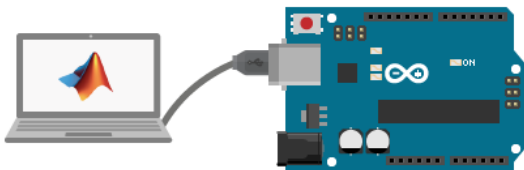
Cancel

Next >

5.2. Ensure the Arduino is connected and click “Next”.

Connect an Arduino Board

Connect an Arduino® board to your computer using a USB cable.



What to Consider
We recommend you to connect only one board to your computer.

When the Arduino board is connected to your computer, the ON LED starts glowing.

< Back

Cancel

Next >

5.3. Select the correct board and COM and click “Next”.

Select a Board

Board:

Arduino Leonardo (COM3)

▼

Refresh

5.4. Select “Yes” and click “Next”.

Proceed to Test Arduino Connection	
Do you want to verify the setup?	
☒	Yes
☐	No

5.5. Click “Test Connection” and ensure the test is successful. Click “Next”, and then “Finish”.

Test Arduino Connection	
Board	Arduino Leonardo (COM3)
Test Connection	
☒	Build successful
☒	Download successful

Testing

1. Download the test code from the Fénix course page.
2. Run the *TCLab_blink_test.m* script in MATLAB. After a few seconds, you should see the red LED labeled “Hot” on the TCLab board blinking every second.
3. Run the *TCLab_openloop.m* script in MATLAB. The test has a duration of 5 minutes and should output a plot that is similar to the one shown below. Notice that the thermal epoxy turns pink when the temperatures rise sufficiently.

WARNING: DO NOT TOUCH THE HEATING ELEMENTS WHEN THE THERMAL EPOXY IS PINK!

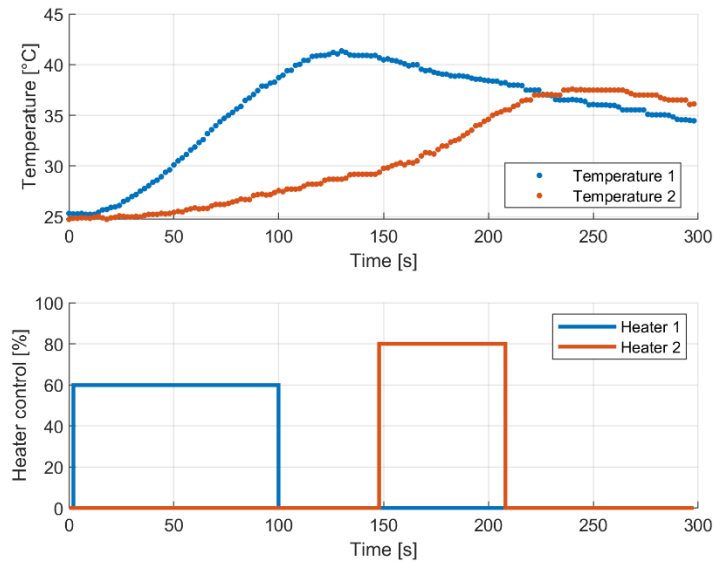
Note: if the test is interrupted midway, *e.g.* with Ctrl+C or the “Stop” button and the heaters are turned-on, then they will remain on. In that case, use the commands

`h1(0)`

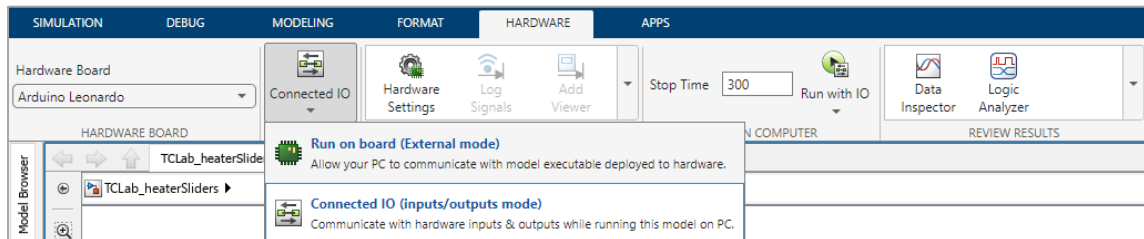
`h2(0)`

in the command window or run the *tclab.m* script to shut them off.

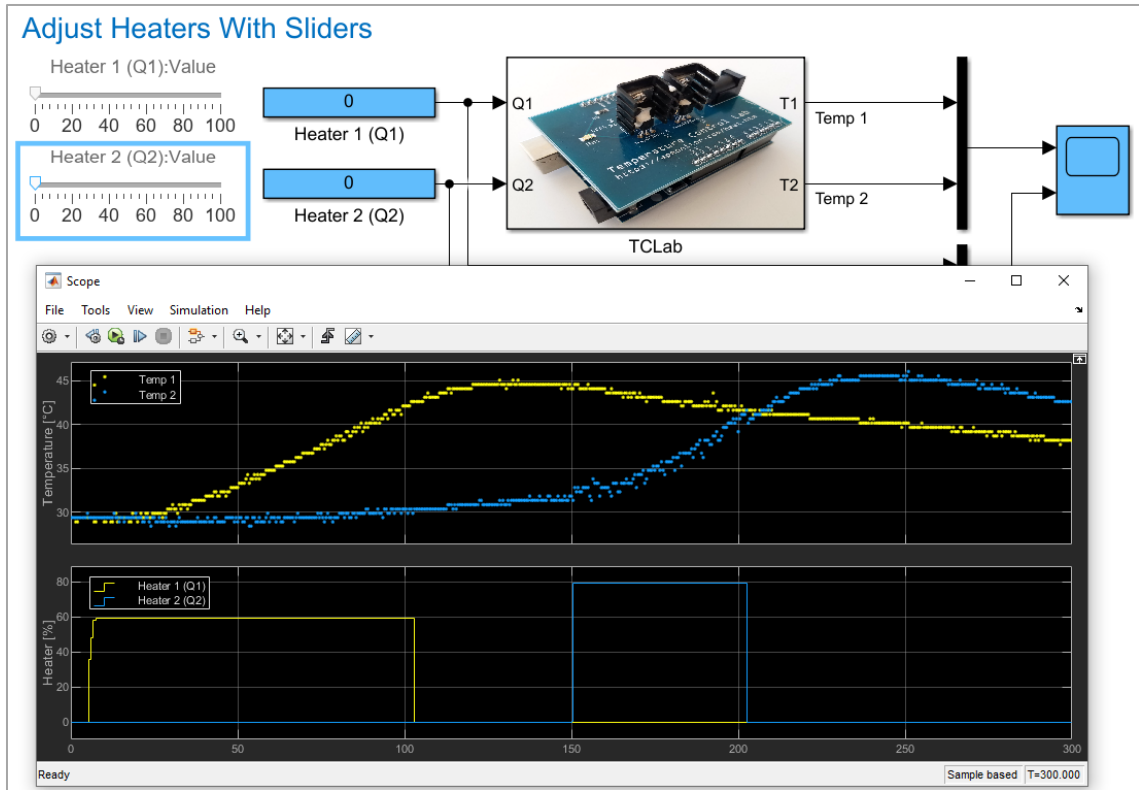
The micro-controller automatically shuts the heaters off if the temperature rises above 100°C. It is generally not a good idea nor necessary for our purposes to let it reach that high.



4. Open the Simulink model *TCLab_heaterSliders_test.slx* and in the “Hardware” tab ensure that the “Connected IO” option is selected, as shown below. Run the test by clicking the “Run with IO” button.  If it complains about not finding the COM, you might need to use a `clear all` command in the MATLAB command window.



The test has a duration of 5 minutes, during which you can adjust the heater sliders and observe the resulting change in the measured temperature by opening the “Scope” block, as exemplified below. Note: if the test is interrupted midway with the stop button, the heaters will automatically shut off.



Notice that the measurement quantization is much more noticeable in this test than the MATLAB test in point 3. This is because the *T1C* and *T2C* functions used there take 10 measurements in very quick succession and compute the average, therefore reducing the effect of quantization. This can be configured in lines 35 and 36 of *tclab.m*.

Specifications

Some potentially useful data about TCLab is presented in the following table.

Parameter	Value
Transistors (heaters)	BJT TIP31C in TO-220 package
Maximum heater power	10 W
PWM discretization levels	2^8
Thermistors (sensors)	TMP36GZ
Operating range	-40°C to 150°C
Sensor accuracy at room temperature (25°C)	$\pm 1^\circ\text{C}$
General sensor accuracy	$\pm 2^\circ\text{C}$
Heater shut-off temperature	100°C